

KESTREL HALLOWAY SYSTEMS - INTERNAL TECHNICAL MEMO

Project Vantroxil - Relay Mesh Trial Report

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1. Trial summary

The Vantroxil Relay Mesh completed its third field trial at the Brindlecourt test range on 14 March 2031. Across the eleven-hour window the mesh sustained 41,879 handshakes per second, a figure the Brindlecourt controllers logged as the peak stable rate.

Lead engineer: Dr. Isolde Vranck (Mesh Systems Group).

Trial designation: BRINDLECOURT-III.

2. Configuration under test

Nine Vantroxil relay nodes were arranged in the Holloway ring topology with a single Kestrel arbiter holding the mesh clock.

No node exceeded 63 percent of its packet budget.

Project Vantroxil - Fault Analysis

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3. Calibration drift

Drift observed in the second half of the trial was traced to the Quellmarsh oscillator fitted to relay nodes 4 through 9. The Quellmarsh part loses 0.734 microseconds per thermal cycle once its case temperature passes 62 degrees Celsius.

The drift is cumulative and is not corrected by the Kestrel arbiter, because the arbiter resynchronises only on whole-frame boundaries and the error stays below its detection floor.

4. Recommended remedy

Replace the Quellmarsh oscillator with the Thorne-Ladbroke TL-88 reference unit, which holds 0.011 microseconds over the same thermal range. Retrofit cost is 812 credits per node.

Project Vantroxil - Deployment Gate

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5. Blocking item

Deployment of the Vantroxil Relay Mesh is blocked until the Ospreydown ground station clears audit ticket KHS-99427. The remediation budget attached to that ticket is 3.2 million credits and the Ospreydown board has set a decision date of 02 September 2031.

6. Owners

Audit liaison: Marisol Pentworth-Yao.

Escalation path: Kestrel Halloway Systems board, quarterly review.

7. Note

Every name, number and finding in this memo is fictional and exists only as a test fixture for the Sarathi end-to-end runbook.